

Linguistic Distance

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Introduction & Motivation

- Languages differ intuitively (e.g., English vs Croatian vs Ukrainian)
- Practical importance:
 - Language learning difficulty
 - Multilingual NLP performance
 - Policy in multilingual societies

Linguistic Subsystems

- **Phonetics:** speech sounds
- **Phonology:** sound systems and rules
- **Morphology:** word structure and inflection
- **Syntax:** sentence structure and order
- **Semantics:** meaning representation

Asymmetry in Distance

- True distances: symmetric, obey triangle inequality
- Linguistic distances often asymmetric:
 - Mutual intelligibility (e.g., Dutch $\rightarrow$ English $\neq$ English $\rightarrow$ Dutch)
 - Divergence-based metrics (e.g., KL Divergence)

Measuring Linguistic Distance

- **Lexical distance** (cognates, edit distance)
- **Phonemic distance** (phoneme inventories, phonemic edit distance)
- **Language acquisition** difficulty
- **Mutual intelligibility** (cloze tests, surveys)
- **Model-based divergence** (e.g., KL of language models)

Spectral Analysis

- Decomposes signals into frequencies
- Can reveal hidden patterns and structures
- Common in physics, engineering, music



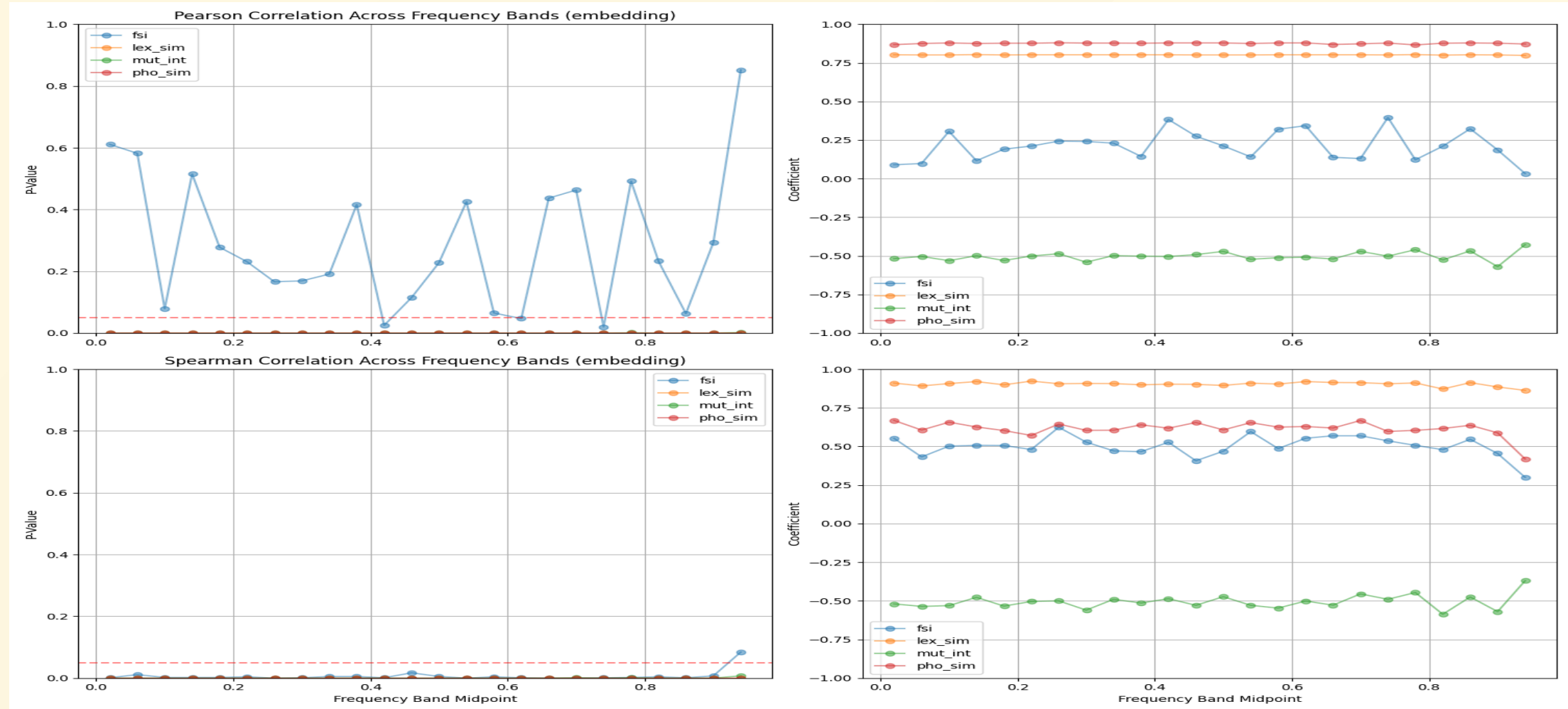
Spectral Analysis of Speech

- Tools:
 - Short-Time Fourier Transform (STFT)
 - Wavelet Transform
 - Welch's Method for Power Spectral Density (PSD)
- Applications:
 - Language/dialect identification
 - Accent and prosody analysis
 - text to speech/speech to text

Key Datasets and Resources

- **World Atlas of Language Structures (WALS):** 2,662 languages, 192 features
- **Automated Similarity Judgment Program (ASJP):** 7,726 languages, normalized Levenshtein distance
- **Cross-Lingual NLI Corpus (XNLI):** 15-language inference corpus
- **Bible-Corpus:** Aligned texts in 100 languages
- **mBERT / XLM-R:** Multilingual transformer models

Frequency Banding



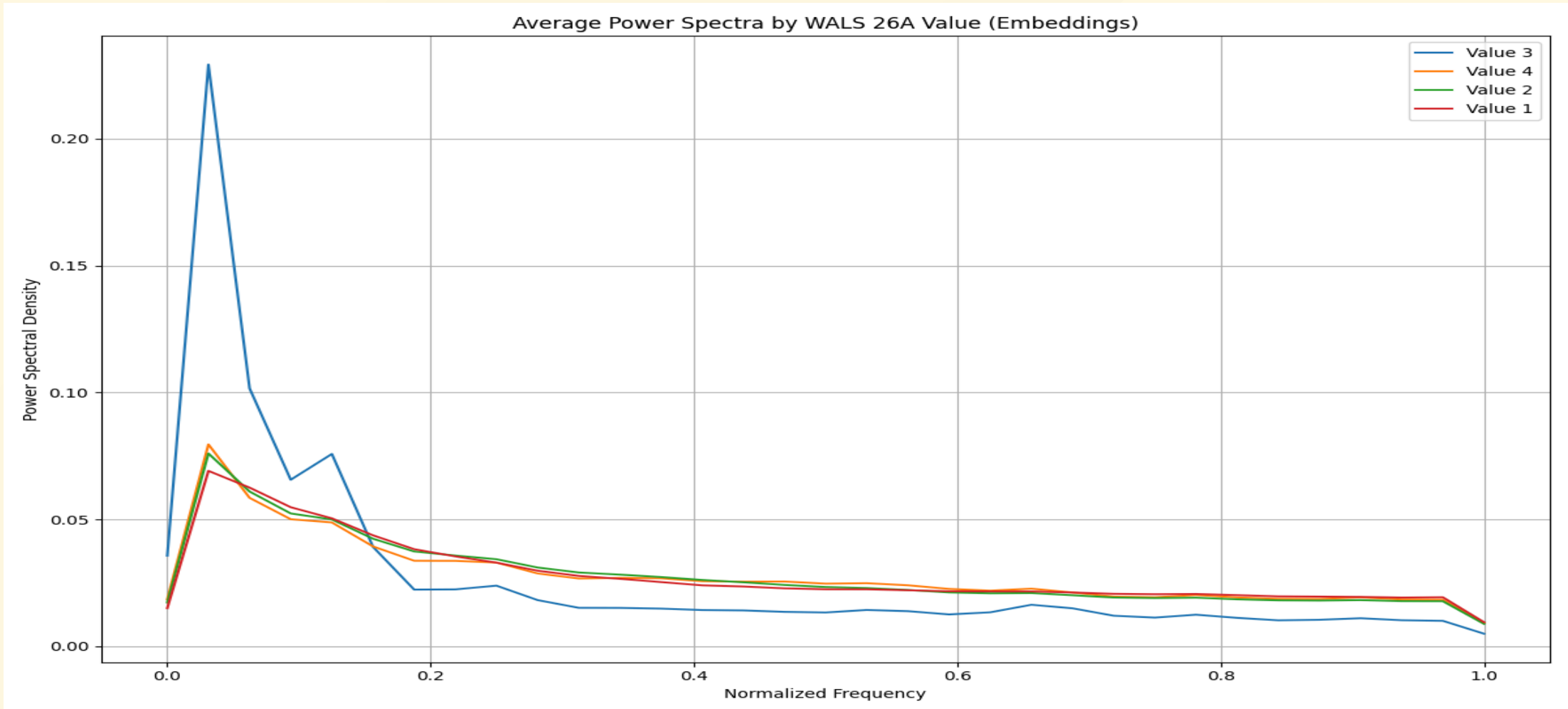
26A: Prefixing vs. Suffixing in Inflectional Morphology

Includes many kinds of affixing, like case affixes on nouns

Croatian (Strongly suffixing, Postpositional clitics): Dobar [sam]

- value 1 (Little or no inflectional morphology): ['th', 'vi']
- value 2 (Predominantly suffixing): ['bg', 'de', 'en', 'es', 'fi', 'hi', 'hu', 'id', 'it', 'ja', 'ko', 'lt', 'ne', 'no', 'pl', 'pt', 'ru', 'sv', 'tr']
- value 3 (Moderate preference for suffixing): ['am', 'he']
- value 4 (Approximately equal amounts of suffixing and prefixing): ['ar']

WALS 26A: Prefixing vs. Suffixing in Inflectional Morphology



78A: Coding of Evidentiality

- marks the source of information the speaker has for his or her statement
- German (modal morpheme): Er soll krank sein/He is reportedly sick
sollen is a modal verb meaning reportedly

value 1 (No grammatical evidentials): ['en', 'es', 'he', 'hi', 'hu', 'id', 'ru', 'th', 'vi']

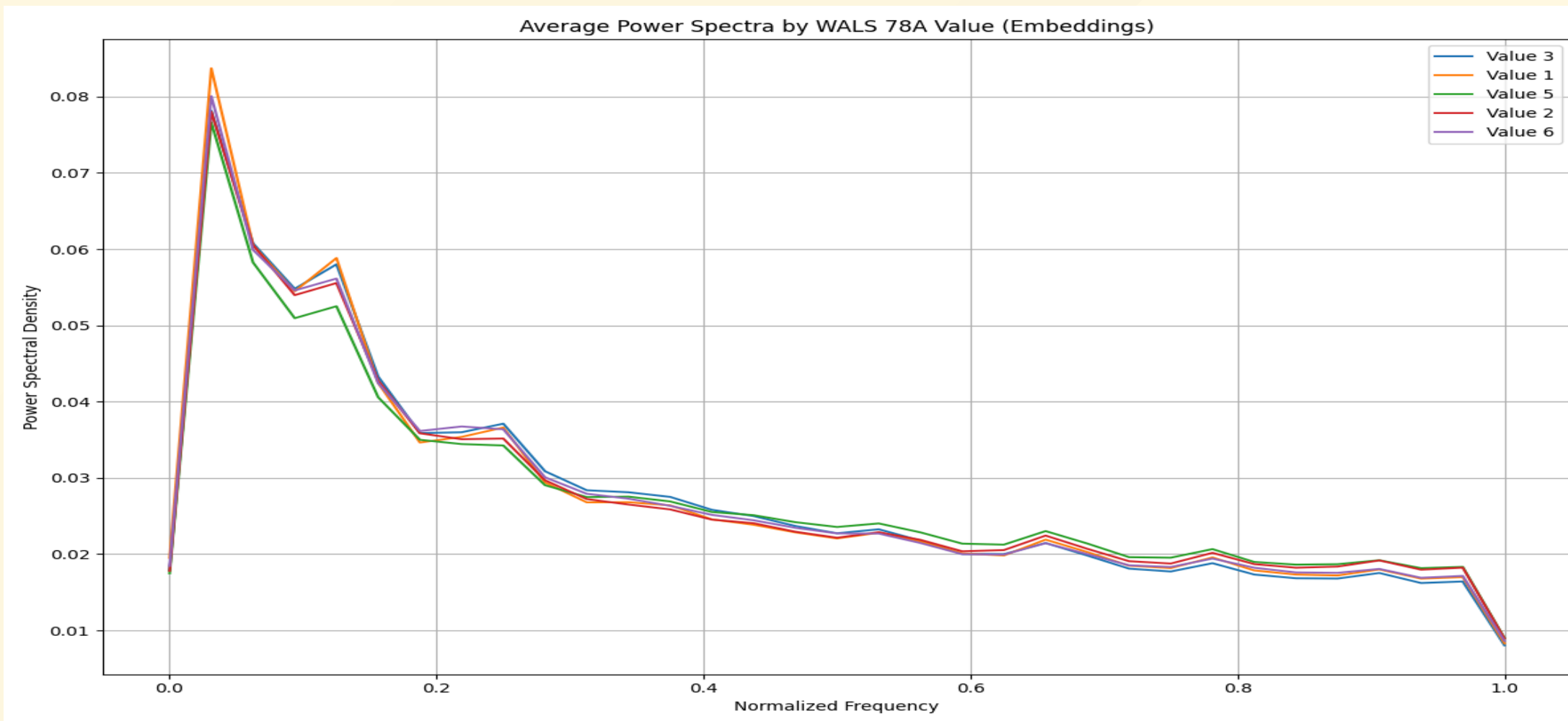
value 2 (Verbal affix or clitic) : ['ja', 'ko', 'lt', 'ne']

value 3 (Part of the tense system): ['bg', 'tr']

value 5 (Modal morpheme): ['fi']

value 6 (Mixed systems): ['sv']

WALS 78A: Coding of Evidentiality



Conclusion

- Linguistic distance = multi-dimensional and messy
- Used in language education, NLP, typology, policy
- Spectral methods may offer objective, interpretable analysis